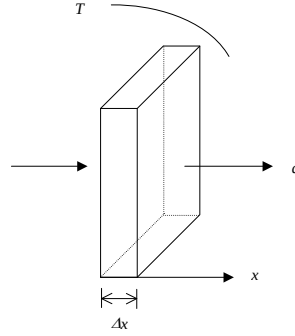


4. Heat Conduction Equation

(1) Derivation of heat diffusion equations



Consider energy balance for a CV of thickness Δx and cross-sectional area A .

$$\dot{E}_{st} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g$$

$$\dot{E}_{st} = mc \frac{\partial T}{\partial t} = \rho c \Delta V \frac{\partial T}{\partial t} = \rho c A \Delta x \frac{\partial T}{\partial t}$$

$$\dot{E}_{in} = q_x = q_x'' A \text{ and } \dot{E}_{out} = q_{x+\Delta x} = q_{x+\Delta x}'' A$$

$$\dot{E}_g = \dot{q} A \Delta x \quad (\dot{q} = \text{volumetric heat generation [W/m}^3])$$

$$\rho A \Delta x \frac{\partial T}{\partial t} = q_x - q_{x+\Delta x} + \dot{q} A \Delta x$$

Taylor series expansion yields $q_{x+\Delta x} = q_x + \frac{\partial q_x}{\partial x} \Delta x + O(\Delta x)^2$

In the above equations, $q_x = -kA \frac{\partial T}{\partial x}$ by Fourier's law

$$\rho c A \Delta x \frac{\partial T}{\partial t} = A \Delta x \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{q} A \Delta x + O(\Delta x)^2$$

$$\text{For } \Delta x \rightarrow 0 \text{ (infinitesimally thin CV), } \rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{q} \quad (1)$$

Similarly, the governing equation in 3-d space can be derived:

$$\boxed{\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q}} \quad (2)$$

Similarly,

$$\text{Cylindrical: } \boxed{\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q}}$$

$$\text{Spherical: } \boxed{\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(kr^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial z} \left(k \sin \theta \frac{\partial T}{\partial z} \right) + \dot{q}}$$

- Temperature profiles formed by heat conduction should always satisfy the above equation at all points. Suppose somebody claims that $T(x, t) = x + t$ is a solution in a particular case where the thermal conductivity is a constant and there is no heat generation within the material. That must be false as the temperature profile cannot satisfy the governing equation.
- The physical meaning of the equation:

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q}$$

$$\Leftrightarrow$$

$$\dot{E}_{st} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g$$

- Special cases

$$1\text{-d: } \rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{q} \quad (3)$$

$$\text{Steady state: } \frac{d}{dx} \left(k \frac{dT}{dx} \right) + \dot{q} = 0 \quad (4)$$

$$\text{Constant thermal conductivity: } \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} + \frac{\dot{q}}{\rho c} \quad (5)$$

$$\text{No heat generation: } \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad (6)$$

(Ex) 1-d, steady-state, constant thermal property, no heat generation: $\frac{d^2 T}{dx^2} = 0 \Rightarrow$ Linear temperature profile

- Thermal diffusivity vs. thermal conductivity:
 - Equation (6) elucidates the physical meaning of thermal diffusivity.
 - In steady heat conduction problems, thermal conductivity is the only parameter.
 - In unsteady problems, thermal diffusivity plays a role that controls the ratio of heat storage to heat diffusion.
 - $\alpha = \frac{\text{Heat diffusion rate}}{\text{Heat storage rate}} \text{ [m}^2\text{/s]}$
 - Consequently, thermal diffusivity determines the temperature change

$$\text{Air } \alpha = \frac{0.0263}{1.1614 \times 1007} = 2.25 \times 10^{-5} \text{ m}^2\text{/s}$$

$$\text{Iron } \alpha = \frac{80.2}{7840 \times 447} = 2.31 \times 10^{-5} \text{ m}^2\text{/s}$$

Larger thermal conductivity means more heat transfer

Larger thermal diffusivity means larger temperature change

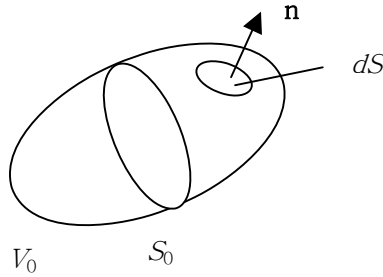
- (Dynamic) viscosity vs. kinematic viscosity
- $\nu \equiv \frac{\mu}{\rho} = \frac{\text{Momentum diffusion rate}}{\text{Momentum storage rate (inertia)}} \text{ [m}^2/\text{s]}$
- Force vs. acceleration
- $a \equiv \frac{F}{m} = \frac{\text{Force}}{\text{Inertia}} \text{ [m/s]}$

(2) Initial and Boundary Conditions

Solutions should satisfy the boundary conditions and initial conditions as well as the governing equation.

- Initial condition: initial temperature distribution, i.e., $T(x, y, z, t = 0) = f(x, y, z)$
- Boundary conditions are assigned from physical observation of the domain boundary
- BC of the 1st kind: known temperature boundary condition, e.g., $T(x = 0, t) = T_0$, phase-change interface, temperature controlled surface
- BC of the 2nd kind: known heat-flux boundary condition, e.g., $q|_{x=0} = q_0''$ or $-k \frac{\partial T}{\partial x} \Big|_{x=0} = q_0''$, insulated surface $\frac{\partial T}{\partial x} \Big|_{x=0} = 0$
- Mixed BC: when a surface is exposed to air and neither temperature nor heat flux is known, $q_{cond}'' = q_{conv}'' + q_{rad}'', \text{ i.e., } -k \frac{\partial T}{\partial x} \Big|_{x=0} = h[T(0) - T_\infty] + \varepsilon \sigma (T^4(0) - T_{sur}^4)$
- Phase-change interface?
- *Boundary conditions are conservation laws applied to the boundary (control surface)!*

(3) Alternative derivation of diffusion equation



Consider the above system of volume V_0 enclosed by S_0 .

Conservation principle:

Change of energy E = Net transfer into the system + generation (sources) – consumption (sinks)

$$\frac{d}{dt} \int_{V_0} E dV = - \int_{S_0} \mathbf{q}'' \cdot \mathbf{n} dS + \int_{V_0} \dot{E}_g dV - \int_{V_0} \dot{E}_c dV,$$

where $\int_{S_0} \mathbf{q}'' \cdot \mathbf{n} dS = \int_{V_0} \nabla \cdot \mathbf{q}'' dV$ by divergence theorem (Note: $\int_{S_0} a_i n_i dS = \int_{V_0} \frac{\partial a_i}{\partial x_i} dV$)

$$\text{Therefore, } \int_{V_0} \left[\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{q}'' + \dot{E}_g - \dot{E}_c \right] dV = 0$$

For the above equation to be valid for an arbitrary volume element,

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{q}'' + \dot{E}_g - \dot{E}_c = 0$$

Since $E = \rho C_p T$ and $\mathbf{q}'' = -k \nabla T$ (Fourier's law),

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{q}'' + \dot{E}_g - \dot{E}_c = 0 \Rightarrow \boxed{\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T)} \text{ Heat Diffusion Equation}$$

Note

- IF $E = \rho$ and $\mathbf{q}'' = \mathbf{j}_a = -\rho D_{ab} \nabla m_a$ (Fick's law), $\boxed{\frac{\partial \rho_a}{\partial t} = \nabla \cdot (\rho D_a \nabla m_a)}$ Mass (Species) Diffusion Equation
- IF $E = \rho u$ and $q'' = \mu \frac{du}{dx}$, $\boxed{\frac{\partial(\rho u)}{\partial t} = \mu \frac{d^2 u}{dx^2}}$ 1-d Momentum Diffusion Equation